

Equivalence Testing for Risk-Compute Engines Beyond Terminal Metrics: Path Functionals and Attribution Summaries*

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Abstract

Institutional risk controls consume more than terminal loss quantiles: investment mandates cap maximum drawdown, committee reporting tracks time underwater and recovery, and remediation decisions are ranked by attribution. Validation practice, however, concentrates on terminal metrics. This paper extends a pre-declared equivalence-testing protocol for risk-compute engines from terminal tail metrics to two further object classes: path functionals (maximum drawdown, time underwater, recovery completion day, jump count) and attribution summaries (portfolio loss, gross attribution, top-family share, attribution concentration). The protocol compares a deterministic low-discrepancy stream against a seeded pseudo-random Monte Carlo reference at 10,000,000 paths per method under acceptance thresholds declared before execution. As instantiated, it is a pre-declared tolerance benchmark rather than a formal inferential equivalence test; the distinction, and the migration path to confidence-bounded margins, are stated explicitly. Two design contributions arise. First, a two-standard split: deterministically replayed quantities (published stress rows) are held to numerical exactness at an absolute tolerance of 10^{-9} , while sampled quantities are held to sampling-scale agreement; mixing the two standards would weaken both. Second, an explicit treatment of integer-valued functionals, whose quantile agreement presents as exact zeros at scale; we argue these are quantization effects, not anomalies—and, conversely, not evidence of distribution-level identity. Across 180 pre-declared checks—96 path-functional rows, 24 terminal-metric cross-check rows, 12 exact stress-replay integrity rows, and 48 stochastic attribution-summary rows—every check passes, with minimum headroom of $1.67\times$ over threshold. The largest gaps concentrate in the empirical-bootstrap family, where inversion of a coarse 199-observation empirical distribution function, not the sampler, drives the difference. All inputs, code revisions, and outputs are sealed under SHA-256 digests, runs are CPU-only with bounded memory, and each experiment is reproducible from two commands when executed against the sealed artifact packs.

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JEL classification: C15; C58; C63; G11; G17

1 Introduction

1.1 The validation gap beyond terminal metrics

A growing class of institutional risk systems replaces or supplements pseudo-random Monte Carlo simulation with deterministic computation: low-discrepancy sampling, quadrature, or closed-form kernels. Whatever the motivation—reproducibility, latency, or audit requirements—any such replacement raises a validation question that is logically prior to questions of model quality: does the deterministic engine compute the same quantities as the stochastic reference it replaces? An earlier paper (Slieptsov, 2026) formalized this question for terminal tail metrics—Value-at-Risk, Conditional Value-at-Risk, and Expected Shortfall—as a pre-declared equivalence test between a deterministic stream and a seeded Monte Carlo reference. A terminal metric depends only on the distribution of portfolio value at the end of the horizon, and this property is what makes it comparatively easy to validate: any method that reproduces the terminal distribution reproduces the metric.

Institutional risk practice does not stop at terminal metrics. Investment mandates cap maximum drawdown. Committee packs report time underwater. Recovery behavior after a trough determines whether a drawdown control releases or stays engaged. Every one of these quantities is a *path functional*: it depends on the order in which returns arrive, not only on where the path ends. Two methods can agree perfectly on the terminal distribution and still disagree on drawdown statistics, because drawdown is computed from the running maximum of the path.

A second class of institutional outputs is the *decomposition*. A portfolio risk number can be right while its attribution is wrong: attribution depends on the joint structure of shocks and weights, not only on the aggregate distribution. Institutional consumers act on the decomposition—concentration committees read top-family share, remediation plans are ranked by contribution, and stress reports assign impacts to named scenarios. If those assignments are not reproducible, the committee is steering on numbers no one can defend. The experiments below test pre-declared *summaries* of that decomposition; the precise scope of what the summaries do and do not establish is stated in Section 5.

This creates a validation gap. A deterministic kernel certified equivalent on VaR and ES has proven nothing about its drawdown, recovery, or attribution numbers. If those numbers drive real controls, they need their own equivalence evidence. This paper supplies it, in the form of two pre-declared experiments executed under a common protocol: Experiment I covers path functionals, and Experiment II covers exact replay of published stress scenarios together with stochastic attribution summaries.

1.2 Contributions

The paper makes four contributions.

First, it extends the Deterministic Risk-Compute Equivalence Test (DRCET) protocol from terminal metrics to two new object classes, publishing the full validation grids, the acceptance thresholds declared before execution, and the complete results: 96 path-functional rows, 24 terminal-metric cross-check rows, 12 exact stress-replay integrity rows, and 48 stochastic attribution-summary rows. All 180 checks pass, with worst-case headroom of $1.67\times$.

Second, it introduces a *two-standard split* for validating attribution surfaces. Attribution reaches a risk report through two computational routes: deterministic scenario replay, which contains no sampling, and stochastic simulation of distributional summaries. The correct standard for the first route is numerical exactness (an absolute tolerance of 10^{-9}); the correct standard for the second is sampling-scale agreement under pre-declared thresholds. Mixing the standards—holding replayed rows to a loose statistical margin, or sampled rows to floating-point exactness—would weaken both. We propose the split as a template for validating any published stress or attribution surface.

Third, it gives an explicit account of how integer-valued path functionals behave under equivalence testing at scale. Time underwater and recovery horizon are day counts; at 10^7 paths their empirical quantiles are pinned to specific integers unless the quantile falls within a small fraction of a path count of a probability-mass boundary. Two well-behaved streams that agree within ordinary sampling error therefore report *identical* integer quantiles almost always. A results column full of exact zeros should be read as quantization, not as suspicious agreement; we state the mechanism and publish the raw metric vectors that allow direct verification. The converse is stated with equal care: identical integer quantiles are not evidence of distribution-level identity.

Fourth, the results locate where genuine sampler-versus-sampler differences live. Across both experiments, smooth transforms of uniforms agree to within 10^{-5} to 3×10^{-4} on continuous functionals, two to three orders of magnitude inside threshold. The only material gaps arise in the empirical-bootstrap family, where both streams invert a 199-observation empirical distribution function; the gap is a property of inverting a coarse step function at scale, is symmetric in interpretation, and grows with horizon. This is an input-data statement, not a sampler statement, and it identifies the empirical-CDF inversion route as the binding constraint on equivalence tightness.

Throughout, the protocol maintains disciplines that are stated once here: acceptance criteria fixed before runs, misses published alongside passes, SHA-256 custody of inputs, code, and outputs, fixed seeds, and explicit claim boundaries. Equivalence between two computational methods is a statement about computation. It is not evidence of forecast validity, not a certification of any production system, and not a claim that either method predicts markets.

2 Related work

Equivalence testing. The statistical logic of demonstrating agreement, rather than failing to demonstrate difference, originates in bioequivalence: the two one-sided tests procedure of Schuirmann (1987) and its intersection–union treatment by Berger and Hsu (1996). The DRCET protocol transports the margin discipline of this literature to computational finance: margins are declared in advance and the burden of proof lies on the claim of agreement. The instantiation in this paper adopts that discipline without the inferential machinery of a formal two-one-sided-tests procedure; Section 3.3 states the distinction precisely and gives the migration path.

Quasi–Monte Carlo. The deterministic stream under test is a scrambled Sobol’ sequence (Sobol’, 1967; Owen, 1998; Joe and Kuo, 2008), a standard low-discrepancy construction whose convergence advantages for financial integrands are surveyed by L’Ecuyer (2009) and Caflisch (1998), and in book form by Glasserman (2004). The present paper is not a convergence study: it asks the narrower operational question of whether a specific deterministic implementation reproduces the statistics of a specific pseudo-random reference (whose generator quality is itself a studied object; Blackman and Vigna, 2018) within margins that institutional consumers would accept.

Backtesting and model risk. Coverage and independence backtests (Kupiec, 1995; Christoffersen, 1998) validate a risk model against realized outcomes. Equivalence testing is orthogonal: it validates one computation against another, holding the model fixed. The two disciplines compose naturally under model-risk governance frameworks such as SR 11-7 (Board of Governors and OCC, 2011) and its April 2026 successor SR 26-2 (Board of Governors, 2026), which call for benchmarking and challenger models; the DRCET protocol can be read as a formalization of the challenger comparison for the compute layer, with pre-declared margins replacing discretion. The pre-registration discipline also responds to the selection and overfitting concerns documented by Lopez de Prado (2018) for research pipelines.

Path functionals and attribution. The distribution of maximum drawdown is analytically nontrivial even for Brownian motion (Magdon-Ismail et al., 2004), and drawdown-based measures appear directly in portfolio optimization (Chekhlov et al., 2005), which is one reason institutional controls consume drawdown statistics rather than terminal quantiles alone. On the attribution side, hot-spot decompositions (Litterman, 1996) and the active-management framework of Grinold and Kahn (2000) established contribution analysis as a committee-facing object; concentration summaries of a decomposition trace back to the Herfindahl–Hirschman index (Hirschman, 1964). Jump components in the stress families used below follow the mixed jump-diffusion tradition of Merton (1976).

3 The DRCET protocol

3.1 Design principles

Both experiments instantiate one protocol, defined by five principles.

- P1. **Pre-declared grid and thresholds.** The full validation grid—object classes, statistics, and acceptance thresholds—is fixed and recorded before execution. The pass criterion cannot be tuned after seeing results. The declaration is internal to the sealed source protocol: no externally timestamped registration exists for these runs, and externally anchored protocol-freeze receipts are adopted for future runs.
- P2. **Two-sided disposition.** Every row is published with status **PASS** (absolute difference within threshold) or **WATCH** (outside threshold). Misses are published alongside passes; the checker additionally treats missing files, malformed contracts, and hash mismatches as validation failures regardless of metric values.
- P3. **Sealed custody.** Input series, code revisions, and outputs are sealed under SHA-256 digests published with the results (Appendix B).
- P4. **Determinism.** Both streams run under fixed seeds; each experiment is reproducible from two commands (Appendix A).
- P5. **Explicit claim boundaries.** The protocol claims computational equivalence within the tested grid, and nothing else: no forecast validity, no production certification, no generalization beyond the tested families and functionals.

3.2 The comparison design

For each test case, two independent streams are generated at 10,000,000 paths per method. The deterministic method is a scrambled Sobol’ stream with a fixed seed, emitted in chunks through the `random()` method of the SciPy Sobol’ generator; Sobol’ uniforms are transformed into returns or shocks by the family-specific map of the experiment. The Monte Carlo reference is a seeded NumPy pseudo-random generator driving the identical map. Because both streams drive the same transform, any disagreement in the resulting metric distributions is attributable to the sampler, which is the object under test.

One construction caveat is disclosed here rather than in a footnote. Because path counts and chunk sizes are operationally arbitrary rather than powers of two, the Sobol’ stream is drawn through repeated `random()` calls, not through `random_base2()`. The runs are fully deterministic and reproducible, but the point set is a chunked Sobol’ stream rather than a pristine power-of-two net, and the star-discrepancy guarantees for complete nets (Owen, 1998; Joe and Kuo, 2008) do not formally apply to it. The equivalence results below stand on measured agreement, not on a discrepancy theorem.

3.3 Threshold construction

The terminal-metric protocol of Slieptsov (2026) binds its equivalence margin to the reference estimator’s own confidence interval, which is the epistemically cleanest construction. The present experiments use fixed pre-declared absolute thresholds instead, for two reasons. First, several functionals under test are integer-valued day counts whose bootstrap quantile confidence intervals collapse to zero width at 10^7 paths, which would make a CI-bounded margin degenerate exactly where the metric is coarsest. Second, fixed thresholds declared before the run preserve the discipline that matters most: the pass criterion cannot be adjusted after results are seen. Migration to CI-bounded margins for the continuous functionals, with an explicit integer-metric rule, is the main methodological item on the roadmap (Section 6).

A second distinction must be stated plainly. As instantiated here, DRCET is a *pre-declared tolerance benchmark*: each comparison is a point distance between one deterministic run and one seeded reference run, checked against a fixed margin. It is not a formal statistical equivalence test in the sense of Schuirmann (1987): no replication across independent scrambles and seeds is performed, no confidence interval for the difference is constructed, and no significance level is attached. Pre-declaration protects against post-hoc tuning; it does not convert a point check into an inferential procedure. The inferential upgrade—independent Sobol’ scrambles paired with independent reference seeds at a reduced path budget, confidence intervals for the paired differences, and CI-inclusion against the declared margins, with the 10^7 -path single-run lane retained as a confirmatory anchor—is the first roadmap item in Section 6.

The margins themselves are operational materiality choices, not derived quantities, and the paper does not claim otherwise. They are deliberately loose for short horizons: 14 days against a 30-day horizon is nearly half the horizon. The protocol’s protection is that they were fixed before execution; attaching a principled basis to each margin—sampling resolution, quantization floor, or control materiality—is part of the CI-bounded migration.

4 Experiment I: Path functionals

4.1 Why path functionals are harder

Three properties make path functionals harder to validate than terminal metrics. First, order dependence: the running maximum is not invariant under permutation of returns, so any sampling scheme that changes the joint ordering behavior of the path can shift drawdown statistics even while preserving each step’s marginal distribution. Second, integer-valued outputs: time underwater and recovery completion day are day counts whose quantiles move in whole-day steps, which changes how agreement and disagreement present themselves in a comparison table (Section 4.8). Third, memory cost: a naive implementation stores the full path matrix; at 10^7 paths and 180 daily steps in double precision that is 14.4 terabytes per case, which is not a practical validation target. The experiment therefore specifies a streaming architecture that carries running state per path and writes only metric vectors to disk-backed

maps; the equivalence claim covers the streaming implementation as run, and the architecture is part of the published method.

4.2 Input data and run contract

The input is a portfolio log-return series of 199 observations, sealed before the run (source file `portfolio_return_series.csv`; its SHA-256 digest is listed in Appendix B). The series is short; that is a disclosed property of the input, not of the test. DRCET measures agreement between two sampling methods driven by the same empirical inputs, so both sides of the comparison inherit the same input limitations symmetrically. Input data quality remains part of the model-risk boundary and is restated in Section 7.

The run contract fixes: 10,000,000 paths per method; chunk size 32,768; horizons of 30, 90, and 180 days; RNG seed 20260714; four path families (below); CPU-only execution on Linux x86_64; Python 3.12.3, NumPy 1.26.4, pandas 2.3.3, SciPy (`scipy.stats.qmc.Sobol`).

4.3 Path families

Four families span the comparison from data-driven to structured dynamics.

Empirical bootstrap. The deterministic stream maps Sobol’ uniforms to empirical quantiles of the observed return sample; the reference stream samples the observed returns with replacement. This family stays closest to the data and makes the fewest assumptions.

Gaussian baseline. Both streams use $\mu + \sigma \Phi^{-1}(u)$, where μ and σ are the empirical mean and standard deviation. The deterministic stream supplies u from Sobol’; the reference supplies Gaussian draws directly. This family isolates the sampler comparison from distributional complexity.

Merton jump proxy. A base normal return plus a Bernoulli jump component in the tradition of Merton (1976). Jump probability is estimated from historical jump observations and capped at 20 percent; jump mean is the mean of negative jump observations, with the 5th percentile as fallback when negative jumps are unavailable; jump standard deviation carries a guarded fallback. This family stresses the comparison with a mixed discrete–continuous distribution.

Two-state regime-switch proxy. Calm and stress volatility states with stress entry probability 0.08 per step and exit probability 0.20 per step; calm σ is $\max(0.65 \sigma, 10^{-5})$ and stress σ is $\max(q_{90}(|r|), 1.8 \sigma)$. Regime entries are counted as the jump-count proxy for this family. Serial state dependence is exactly where path functionals are most sensitive, which is why the family is included.

4.4 Streaming architecture

The builder processes paths in chunks of 32,768. For each chunk it maintains running NAV, running peak, drawdown state, underwater day counts, and recovery bookkeeping, then discards the chunk’s return data. Only the five metric values per path are appended to disk-backed

memory maps; peak resident memory is bounded by the chunk, not by the path count. The published wall time for the complete 24-case run (4 families $\times$ 3 horizons $\times$ 2 streams) is 3,706.476 seconds, about 61.8 minutes, on commodity CPU hardware.

4.5 Functional definitions

For each simulated log-return path $(r_1, \dots, r_T)$ over horizon T , with $\text{NAV}_t = \exp(\sum_{s \leq t} r_s)$ and running peak $P_t = \max\{1, \text{NAV}_1, \dots, \text{NAV}_t\}$:

$$\begin{aligned}
D_t &= \max(0, 1 - \text{NAV}_t/P_t) && \text{(drawdown loss at day } t) \\
\text{TL} &= \max(0, 1 - \text{NAV}_T) && \text{(terminal loss)} \\
\text{MDD} &= \max_{t \leq T} D_t && \text{(maximum drawdown loss)} \\
\text{TUW} &= \#\{t \leq T : D_t > 10^{-6}\} && \text{(time underwater, days)} \\
\text{REC} &= \min\{t > t^* : \text{NAV}_t \geq P_{t^*}(1 - 10^{-9})\} \text{ else } T+1 && \text{(recovery completion day, censored)} \\
\text{JC} &= \text{path jump-mask sum, or regime-entry count} && \text{(jump count)}
\end{aligned}$$

where t^* is the day of the maximum drawdown trough. Three definitional choices deserve comment. First, the recovery functional records the completion *day index* within the path, not the elapsed duration since the trough: a path whose trough falls on day t^* and which first regains the trough-time peak on day t reports the value t itself. The day-count thresholds of Table 1 are declared on this day-index scale, and the censoring point $T+1$ is accordingly tied to the horizon rather than to the trough date. A duration-based variant ($t - t^*$, censored at $T - t^* + 1$) is a different functional and is not the object tested here; conflating the two would change the meaning of the quantiles and thresholds. Second, a path that never recovers within the horizon is recorded at $T+1$ rather than dropped, so the recovery distribution carries no silent missing mass. Third, the underwater tolerance of 10^{-6} prevents floating-point dust at the peak from being counted as underwater days. All choices apply identically to both streams, so none biases the comparison; tie-breaking between equal maximum-drawdown troughs follows the sealed implementation. Terminal loss is retained in the grid even though it is a terminal metric already covered by Slieptsov (2026); it serves as an internal cross-check—if the two streams disagreed on terminal loss, any agreement on path functionals would be suspect.

4.6 Validation grid and thresholds

The grid is 4 families $\times$ 3 horizons $\times$ 5 metrics $\times$ 2 statistics = 120 rows. Statistics are p_{95} and p_{99} for the four loss and duration metrics, and mean and p_{95} for jump count. All thresholds were declared before the run as absolute differences between the deterministic statistic and the Monte Carlo reference statistic (Table 1); the status rule is **PASS** if the absolute difference is at or below threshold, **WATCH** otherwise.

Table 1: Experiment I: pre-declared acceptance thresholds (absolute differences).

Metric	Statistic	Threshold
terminal loss	p_{95}	0.015 (1.5 pp)
terminal loss	p_{99}	0.025 (2.5 pp)
maximum drawdown loss	p_{95}	0.015 (1.5 pp)
maximum drawdown loss	p_{99}	0.025 (2.5 pp)
time underwater	p_{95}	7 days
time underwater	p_{99}	14 days
recovery (or $T+1$)	p_{95}	7 days
recovery (or $T+1$)	p_{99}	14 days
jump count	mean	0.35
jump count	p_{95}	1.0

4.7 Results

The headline result is 120 PASS, 0 WATCH, 0 FAIL. Table 2 lists the five largest absolute differences across the grid, all under the empirical bootstrap or the Merton proxy at the 180-day horizon; Table 3 reports the per-case summary. The largest observed gap in any tail statistic is 0.0100 absolute, on maximum drawdown p_{99} under empirical bootstrap at 180 days, against a threshold of 0.025.

Table 2: Experiment I: five largest absolute differences across the 120-row grid.

Case	Horizon	Metric	Stat.	Determ.	MC ref.	Abs. diff	Thresh.	Headroom
emp. bootstrap	180	max drawdown	p_{95}	0.305865	0.314855	0.008990	0.015	1.67×
emp. bootstrap	180	terminal loss	p_{95}	0.236748	0.245089	0.008341	0.015	1.80×
emp. bootstrap	180	max drawdown	p_{99}	0.372110	0.382114	0.010005	0.025	2.50×
emp. bootstrap	180	terminal loss	p_{99}	0.324101	0.333928	0.009827	0.025	2.54×
Merton proxy	180	jump count	mean	9.949748	9.951426	0.001678	0.35	208.63×

4.8 Exact zeros on integer functionals are expected, not suspicious

Time underwater and recovery are day counts. Their p_{95} and p_{99} are order statistics of integer-valued distributions, and at 10,000,000 paths the empirical quantile of such a distribution is pinned to a specific integer unless the quantile falls within a fraction of a path count of a probability-mass boundary. Two well-behaved streams that agree on the underlying distribution to within ordinary sampling error will therefore report identical integer quantiles almost always. The exact zeros in Table 3 are the discrete analogue of the sub-basis-point agreement on the continuous rows. A reviewer who sees a column of zeros should read it as quantization, not as copied numbers; the raw metric vectors in the sealed pack allow direct verification.

The converse limitation is stated for completeness: identical integer quantiles do not imply identical distributions. Two streams could differ in censoring mass at $T+1$, in median duration, or elsewhere in the CDF while agreeing at p_{95} and p_{99} . Distribution-level comparisons for the

Table 3: Experiment I: per-case wall time and absolute differences (selected columns).

Family	Horizon	Time, s	TL p_{95}	TL p_{99}	MDD p_{95}	MDD p_{99}	JC mean
emp. bootstrap	30	66.7	0.0035356	0.0048217	0.0040945	0.0050828	0.000000
emp. bootstrap	90	163.5	0.0061933	0.0077073	0.0068518	0.0081393	0.000000
emp. bootstrap	180	385.0	0.0083414	0.0098271	0.0089899	0.0100045	0.000000
Gaussian	30	70.0	0.0000359	0.0000826	0.0000423	0.0001459	0.000000
Gaussian	90	177.5	0.0001167	0.0001269	0.0000593	0.0001505	0.000000
Gaussian	180	400.4	0.0000303	0.0001018	0.0000386	0.0000015	0.000000
Merton proxy	30	122.9	0.0000157	0.0000799	0.0000219	0.0000046	0.000117
Merton proxy	90	333.0	0.0000861	0.0001467	0.0001245	0.0002541	0.000144
Merton proxy	180	811.4	0.0000154	0.0001690	0.0000399	0.0000135	0.001678
regime switch	30	119.6	0.0000608	0.0002267	0.0000320	0.0001656	0.000268
regime switch	90	306.3	0.0000121	0.0000875	0.0000749	0.0000155	0.000180
regime switch	180	750.0	0.0000523	0.0001429	0.0000871	0.0001666	0.000085

Time-underwater and recovery differences were exact zero on every row of the results table; Section 4.8 explains why exact agreement is the expected behavior for integer-valued functionals at this sample size.

day-count functionals—censoring fraction, maximum CDF distance, and survival-curve distances computed from the sealed per-path vectors—are specified as a post-processing addendum on the roadmap; they require no new simulation.

4.9 Where the gaps live and why

The parametric families agree to within a few basis points at every horizon: Gaussian, Merton, and regime-switch rows sit at absolute differences of 10^{-5} to 3×10^{-4} on the continuous functionals, two to three orders of magnitude inside threshold. This is the expected behavior of a low-discrepancy stream against pseudo-random sampling when the transform from uniforms to returns is smooth (L’Ecuyer, 2009; Glasserman, 2004).

The empirical bootstrap family carries every one of the largest gaps, and the gaps grow with horizon: roughly 0.004 at 30 days, 0.007 at 90 days, 0.010 at 180 days on the drawdown and terminal quantiles. The dominant mechanism is the transform, not the sampler drifting apart from the model. The empirical quantile function of a 199-observation sample is a step function with 199 flat segments. A stratified low-discrepancy stream and a pseudo-random stream allocate mass across those steps differently, and the difference compounds over more steps at longer horizons. The gap arises from the interaction between the discrete empirical-quantile transform and the two sampling schemes, is symmetric in interpretation, and stays inside the pre-declared thresholds with at least $1.67\times$ headroom. A longer input series is expected to shrink it, though the dependence on sample depth remains to be measured across input panels (Section 6); the statement concerns the input layer, not the sampler.

5 Experiment II: Stress replay and attribution summaries

5.1 Two computation modes, two standards

Attribution reaches a risk report through two different computational routes, and a credible test must cover both.

The first route is deterministic scenario replay. A published stress row is, mechanically, a sum: exposed weight times shock. There is no sampling in it. The correct standard for this route is exactness. If an engine cannot reproduce its own published stress rows from source weights and shock definitions to numerical precision, nothing downstream deserves trust. Part one of the experiment tests this route at an absolute tolerance of 10^{-9} ; the tolerance is deliberate, because stress replay contains no sampling, so any deviation beyond floating-point accumulation indicates a defect in weights, shock definitions, or bookkeeping, and the test is designed so that such a defect cannot hide inside a loose tolerance.

The second route is stochastic. Distributional attribution summaries—tail quantiles of the portfolio loss ratio, or the concentration of attribution across families—are estimated by simulation. Here the correct standard is the DRCET equivalence standard of Section 3.2: a deterministic low-discrepancy stream must agree with a seeded Monte Carlo reference within pre-declared thresholds at a fixed path budget. Part two tests this route at 10,000,000 paths per method.

Splitting the test this way keeps the standards honest: exact things are held to exactness, sampled things are held to sampling-scale agreement. Mixing the two standards in one criterion would weaken both.

5.2 Source data and exposure map

The source is a sealed factor-attribution completion pack (directory `factor_attribution_v1_completion_20260615`) containing portfolio factor exposures, stress scenario definitions, a published factor universe of 26 factors, and per-asset coverage files. The portfolio maps to seven dominant exposure families (Table 4); these are dominant sleeve weight shares from a free-data proxy factor model, not vendor factor betas, and no equivalence to commercial production attribution (Barra, Axioma, MSCI) is claimed.

Table 4: Experiment II: dominant-family exposure map (normalized weight shares).

Dominant family	Weight share	Symbols
technology growth equity	0.5036996672	AAPL, NVDA, MSFT
communication services equity	0.4205953601	META
commodity proxy (limited)	0.0345744064	USO, SLV, CPER
commodity (mapping review)	0.0195364795	GOLD
crypto market	0.0127453141	BTC, XRP, STRK, ADA, ETH, SOL
financial equity	0.0073689234	JPM
cash settlement	0.0014798493	USDT

The run contract fixes: 10,000,000 paths per stochastic method; chunk size 262,144; four stochastic profiles (Section 5.4); six stress scenarios; RNG seed 20260714; CPU-only Linux x86_64; the same software stack as Experiment I.

5.3 Part one: exact stress replay

For each published stress scenario, the computed exposed weight is the sum of normalized weights of the families named in the scenario, and the replayed NAV impact is the product of the shock and the exposed weight. The row passes if both the exposure difference and the impact difference against the published values are at or below 10^{-9} in absolute value. Table 5 reports the scenario set and results: all six scenarios reproduced published impacts exactly or to machine epsilon, with the largest absolute deviation at 4.94×10^{-17} . The non-zero entries are single floating-point rounding steps at double precision—the expected residue of summation order. Nothing in the replay set reached even 10^{-15} of the 10^{-9} tolerance.

Table 5: Experiment II, part one: exact stress replay at tolerance 10^{-9} .

Scenario	Shock	Exposed weight	Replayed impact	Abs. diff
US mega-cap technology shock	−20.00%	92.4295%	−18.4859%	0
Communication services de-rating	−18.00%	42.0595%	−7.5707%	2.78×10^{-17}
Technology hardware drawdown	−22.00%	50.3700%	−11.0814%	0
Commodity sleeve shock	−15.00%	5.4111%	−0.8117%	1.39×10^{-17}
Crypto market crash	−50.00%	1.2745%	−0.6373%	4.94×10^{-17}
Settlement sleeve par shock	−2.00%	0.1480%	−0.0030%	3.59×10^{-19}

Each row is checked on two differences separately—exposed weight and NAV impact—both at the 10^{-9} tolerance; the table reports the impact difference. Printed values are rounded for display; the comparison runs on the full-precision sealed inputs, which is why residues at the 10^{-17} scale are resolvable.

Part one establishes that the published stress rows are mechanically reproducible from their declared inputs to machine precision. This is a low bar stated precisely, and stating it precisely is the point: an attribution stack whose published rows cannot be replayed exactly has a bookkeeping defect that no distributional testing can compensate for. Passing at 10^{-9} removes that entire defect class from the table before any statistics are discussed. The classification of the evidence should be equally precise: part one is an *integrity* check of the published surface against its declared inputs, not sampler-equivalence evidence; the equivalence content of the experiment resides entirely in part two. The experiment’s headline is accordingly read as 12 integrity rows plus 48 equivalence rows, not as 60 rows of one kind.

5.4 Part two: stochastic attribution profiles

The stochastic stream simulates family-level shocks with one common factor and idiosyncratic family shocks:

$$\text{shock}_i = \sigma_i \left(\rho z_{\text{common}} + \sqrt{1 - \rho^2} z_{\text{idio},i} \right), \quad (1)$$

where the deterministic method supplies the z -values through a scrambled Sobol’ stream with a fixed seed and the Monte Carlo reference supplies them through a seeded NumPy generator. Both streams drive the identical shock model, so any disagreement in the metric distributions is attributable to the sampler. The chunked-Sobol’ caveat of Section 3.2 applies unchanged.

Four profiles vary the common-factor loading and switch stress and jump overlays on and off: a balanced factor profile ($\rho = 0.55$, no overlays), a cluster-stress profile ($\rho = 0.70$, stress overlay on), a jump profile ($\rho = 0.45$, jump overlay on), and a detoned residual profile ($\rho = 0.15$, no overlays). One naming precision matters: in construction (1), ρ is the *loading* on the common factor, so the implied pairwise correlation between two family shocks is ρ^2 —0.49 at the highest profile value, not 0.70. The source packs label the parameter loosely as a correlation; the economic reading should follow the loading semantics stated here. The distinction does not affect the equivalence comparison, because both streams drive the identical construction. Family volatility proxies are: cash settlement 0.005; crypto $\max(0.18, |\text{shock}|/1.75)$; commodity families $\max(0.07, |\text{shock}|/1.75)$; financial equity 0.085; other equity families $\max(0.08, |\text{shock}|/1.9)$. The cluster-stress overlay applies a stress mask with probability 0.12 and volatility multipliers of 2.40 (technology), 2.25 (communication), 1.35 (financial), 1.55 (crypto), and 0.05 (cash). The jump overlay uses, per family: crypto (probability 0.065, mean -2.2σ , s.d. 0.80σ); technology/communication (0.045, -1.8σ , 0.55σ); commodity (0.035, -1.4σ , 0.55σ); cash (0.005, -0.02 , 0.005); other (0.030, -1.3σ , 0.45σ). These parameters define proxy stress mechanics for the equivalence comparison; they are applied identically to both streams and are not presented as calibrated market dynamics.

5.5 Attribution metric definitions

For family shocks s_i and normalized family weights w_i , with contributions $c_i = s_i w_i$, total $\tau = \sum_i c_i$, gross $g = \sum_i |c_i|$, and shares $\pi_i = |c_i|/\max(g, \varepsilon)$:

$$\begin{aligned} \text{portfolio loss ratio} &= \max(0, -\tau) \\ \text{gross attribution} &= g \\ \text{top-family share} &= \max_i \pi_i \\ \text{attribution HHI} &= \sum_i \pi_i^2 \\ \text{technology-common loss ratio} &= \max(0, -\sum_{i \in \text{tech} \cup \text{comm}} c_i) \\ \text{limited-mapping loss ratio} &= \max(0, -\sum_{i \in \text{commodity} \cup \text{cash} \cup \text{review}} c_i) \end{aligned}$$

The six metrics cover the three questions a committee asks of an attribution: how much is lost in total; how concentrated the loss drivers are (top-family share and HHI, in the sense of Hirschman, 1964); and whether the loss travels through the dominant channel or through sleeves with limited mapping quality. The last metric exists specifically so that mapping-quality risk is measured rather than absorbed silently into a residual.

One scope statement belongs here rather than in the limitations. All six metrics are invariant

under relabelings of the contribution vector that preserve the two named groups: total, gross, largest share, and share concentration are invariant under any permutation of $(c_1, \dots, c_7)$, and the two group sums are invariant under permutations inside their groups. The stochastic grid therefore validates the distributional *summaries* of the decomposition that committees consume, not the identity of each named contribution: a within-group relabeling that preserved these summaries would pass every stochastic row. Equivalence of the named contribution vector itself requires per-family rows—signed contribution quantiles, top-family identity agreement, rank agreement—which need per-family metric vectors that the present packs do not retain; they are specified on the roadmap, and the paper’s claims are scoped to attribution summaries throughout.

5.6 Validation grid, thresholds, and results

The grid is 6 scenarios $\times$ 2 exact metrics = 12 exact replay rows, plus 4 profiles $\times$ 6 metrics $\times$ 2 statistics (p_{95} , p_{99}) = 48 stochastic rows; 60 rows in total. Pre-declared stochastic thresholds are 0.0025 (p_{95}) and 0.005 (p_{99}) for the four loss-ratio metrics, and 0.010 (p_{95}) and 0.0175 (p_{99}) for top-family share and attribution HHI; replay rows carry the 10^{-9} tolerance on both exposed weight and impact.

The headline result is 60 PASS, 0 WATCH, 0 FAIL. Table 6 lists the largest absolute gaps across the 48 stochastic rows. The largest stochastic gap is 0.000253 absolute, on attribution HHI p_{95} under the cluster-stress profile, against a threshold of 0.010—a headroom of 39.5 $\times$. The tightest utilization on the tightest metric family is the cluster-stress portfolio loss ratio at p_{95} : deterministic 0.2736883208 versus reference 0.2737791613, an absolute difference of 0.0000908405 against a threshold of 0.0025, headroom 27.5 $\times$. Per-profile wall times were 35.20 s (balanced), 37.94 s (cluster stress), 50.72 s (jump), and 44.05 s (detoned residual); the full pack completed in 167.987 seconds on commodity CPU hardware. The contrast with Experiment I’s 3,706 seconds reflects the object dimensions: path functionals carry per-step state over up to 180 daily steps, while attribution draws one cross-sectional shock vector of seven families per path.

Table 6: Experiment II, part two: largest absolute gaps across the 48 stochastic rows.

Profile	Metric	Stat.	Determ.	MC ref.	Abs. diff	Thresh.	Headroom
cluster stress	attribution HHI	p_{95}	0.752522	0.752269	0.000253	0.010	39.5 $\times$
balanced factor	attribution HHI	p_{95}	0.769794	0.770018	0.000224	0.010	44.7 $\times$
jump factor	gross attribution	p_{99}	0.243388	0.243569	0.000181	0.005	27.6 $\times$
cluster stress	gross attribution	p_{99}	0.380987	0.381163	0.000177	0.005	28.3 $\times$
detoned residual	gross attribution	p_{99}	0.212644	0.212780	0.000136	0.005	36.9 $\times$

5.7 Interpretation

The stochastic gaps are one to two orders of magnitude smaller than the empirical-bootstrap gaps of Experiment I, which is expected: the attribution transform (1) is smooth in its uniforms, there is no coarse empirical quantile inversion in the loop, and the cross-sectional dimension is seven rather than up to 180 serial steps. The attribution HHI levels themselves (about 0.75 to 0.77 at p_{95} across profiles) reflect the portfolio in the source pack, which is dominated by two equity families totaling over 92 percent of weight. The test does not judge that concentration; it verifies that both streams measure the same concentration to within 2.5 basis points of share. The distinction matters for reading the tables: the experiment validates the ruler, not the portfolio—and, per the scope statement of the preceding subsection, the summaries, not the labels.

Agreement between two implementations of a proxy shock model does not make the proxy model a production factor model, and the paper claims no such thing. The profile parameters are fixed mechanics for the comparison, not calibrated dynamics. Attribution equivalence also says nothing about forecast quality or allocation merit.

6 Discussion

6.1 What the results establish

Taken together, the two experiments establish that, for the tested families, horizons, profiles, and inputs, a deterministic low-discrepancy CPU stream reproduces Monte Carlo reference statistics for path functionals and attribution summaries within thresholds declared before execution, at a 10^7 -path budget, in bounded memory—and that published stress rows replay exactly at 10^{-9} . The benchmark discipline that Slieptsov (2026) established for terminal metrics therefore extends to the two object classes institutional drawdown controls and attribution reports actually consume. Table 7 states the composition of the 180 checks and the claim boundary of each block.

Table 7: Overview of the 180 pre-declared checks.

Object class	Rows	Standard	Result	Claim boundary
Terminal-metric cross-check	24	fixed tolerance	all pass	internal consistency
Path functionals	96	fixed tolerance	all pass	sampler agreement
Exact stress replay	12	10^{-9} exactness	all pass	bookkeeping integrity
Attribution summaries	48	fixed tolerance	all pass	summaries, not named vector

The comparative anatomy of the gaps is itself informative. Sampler-versus-sampler differences on smooth transforms are negligible at this budget; the binding constraint on equivalence tightness is the empirical-CDF inversion route, where input coarseness (199 observations) dominates. For practitioners, the implication is direct: when a deterministic engine and a

Monte Carlo engine disagree materially on drawdown statistics, the first suspect should be the empirical input layer and its quantile inversion, not the low-discrepancy sampler.

6.2 What the results do not establish

The results do not establish forecast validity—both experiments test whether two computational methods agree with each other, not whether either predicts markets. They do not certify any production system: equivalence evidence is necessary but not sufficient for production deployment. They do not constitute a formal inferential equivalence test (Section 3.3), and within Experiment II they validate attribution summaries, not the identity of the named contribution vector. They do not generalize automatically to path families, functionals, or attribution surfaces outside the tested grids, and the deterministic method under test is a specific Sobol’ streaming harness, not every deterministic method for every path-dependent model.

6.3 Methodological roadmap

Six items follow directly. First, the inferential upgrade of Section 3.3: independent Sobol’ scrambles paired with independent reference seeds at a reduced fixed path budget, confidence intervals for the paired differences, CI-inclusion against declared margins, and a published integer-metric rule for day counts—with the 10^7 -path single-run lane retained as a confirmatory anchor, unifying the threshold construction with Slieptsov (2026). Second, vector-level attribution rows: per-family signed contribution quantiles, top-family identity agreement, rank and sign agreement, closing the gap between summary equivalence and named-decomposition equivalence. Third, correctness controls that sampler equivalence cannot supply by construction: known-answer fixtures with hand-computable paths, an independent reimplement of the functionals, and deliberate negative controls—shuffled return order (which must move drawdown statistics while nearly preserving terminal loss), family label swaps (which the vector-level rows must fail), and 10^{-8} exposure perturbations (which the exact-replay tolerance must fail). Fourth, distribution-level comparisons for the day-count functionals—censoring mass at $T+1$, CDF and survival-curve distances—computable from the sealed per-path vectors without new simulation. Fifth, cross-substrate repeats (GPU, ARM) under the same grids and thresholds. Sixth, longer and multi-asset input panels, so the empirical-bootstrap transform gap can be measured as a function of sample depth; production vendor exposures for the attribution grid where licensing permits; additional functionals with disclosed definitions (Calmar-style ratios, conditional drawdown (Chekhlov et al., 2005), second-drawdown statistics); and externally timestamped protocol-freeze receipts for future runs. An external submission protocol, so third parties can run the same grids against their own kernels and publish comparable rows, is specified in the terminal-metric paper and applies here without modification.

7 Limitations

Six limitations bound the claims. (i) *Input coarseness*. The path-functional input is a single 199-observation return series; the empirical-bootstrap gaps in Section 4.9 partly reflect that coarseness, and source data quality remains inside the model-risk boundary of the platform that produced the source packs. (ii) *Proxy factor model*. The attribution exposures are dominant-family weight shares from a free-data model with 26 published factors and 7 dominant families; they are not vendor production attribution, and no vendor equivalence is claimed. (iii) *Fixed shock mechanics*. Profile correlations, volatility proxies, and overlay parameters are declared constants for the equivalence comparison, not estimates of market dynamics. (iv) *Threshold construction*. Thresholds are fixed and pre-declared rather than CI-bounded (Section 3.3). (v) *Sobol’ construction*. The deterministic stream is a chunked `random()` stream, not a power-of-two net; discrepancy-theoretic guarantees do not formally apply, and the claim rests on measured agreement. (vi) *Substrate coverage*. All runs are CPU-only; cross-substrate equivalence is a roadmap item, not a claim of this paper. (vii) *Shared transform*. Both streams share the family maps, functional definitions, and attribution code; the comparison isolates the sampler by construction, which equally means that a defect common to the shared code would reproduce identically on both sides and pass. Known-answer fixtures and an independent reimplementations of the functionals are the appropriate controls; both are roadmap items, and neither is claimed here.

8 Conclusion

Validation practice for risk computation has concentrated on the object class that is easiest to test. This paper moved the benchmark discipline to the two classes that institutional controls actually consume—trajectory statistics and attribution summaries—and proposed a standard for each: sampling-scale agreement under pre-declared thresholds for simulated quantities, and numerical exactness for replayed ones. On a 180-check combined grid at ten million paths per method, the deterministic stream under test met both standards everywhere, with the binding gaps traced to input coarseness rather than sampler behavior. The protocol’s disciplines—pre-declared acceptance criteria, published misses, sealed custody, fixed seeds, explicit claim boundaries—are independent of any particular engine, and the author’s hope is that the grids published here are run against other kernels, by other parties, with comparable rows published either way.

Reproducibility and data availability

Each experiment reproduces from two commands (a builder and a checker; Appendix A) when executed against the sealed artifact packs, on commodity CPU hardware: Linux x86_64, Python 3.12.3, NumPy 1.26.4, pandas 2.3.3, SciPy (`scipy.stats.qmc.Sobol`); RNG seed 20260714

throughout; wall times 3,706.476 s (Experiment I) and 167.987 s (Experiment II). The sealed artifact packs—raw per-path metric vectors, result grids, run logs, and manifests—are identified by the SHA-256 digests in Appendix B and are available from the author upon request. The git revision of the deployed runtime was not available in the deployed-copy environment and is disclosed as absent rather than substituted. The same disclosure discipline applies to items the run manifests did not capture: the exact SciPy version, the NumPy bit-generator identifier, and the CPU model and memory of the host were not recorded, and are reported as absent rather than reconstructed.

Declarations

The author is the founder and chief executive officer of Scientific Analytics Alliance Inc., which develops the software harness under test; readers should weigh the equivalence evidence accordingly, which is one reason every acceptance threshold was declared before execution and every artifact is hash-sealed. The author previously served as Head of Interbank Cash Operations and of the NBU Cash Transactions Department at PrivatBank JSC CB. This research received no external funding. The paper is descriptive computational research; it is not investment advice and contains no trading recommendation.

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A Reproduction commands and environment

Experiment I (path functionals):

```
python3 scripts/build_drcet2_path_functional_10m_cpu_stream.py \
    -paths 10000000 -chunk-size 32768 -horizons 30,90,180 \
    -cases empirical_bootstrap,gaussian_baseline,
        merton_jump_proxy,regime_switch_proxy
python3 scripts/check_drcet2_path_functional_10m_cpu_stream.py
```

Checker output (abridged): status PASS; 10,000,000 paths per method; 120 rows; metrics: jump count, maximum drawdown loss, recovery days (or horizon+1), terminal loss, time underwater days.

Experiment II (attribution):

```
python3 scripts/build_drcet3_attribution_10m_cpu_stream.py \
    -paths 10000000 -chunk-size 262144 \
    -profiles balanced_factor_proxy,cluster_stress_proxy,
        jump_factor_proxy,detoned_residual_proxy
python3 scripts/check_drcet3_attribution_10m_cpu_stream.py
```

Checker output (abridged): status PASS; 10,000,000 paths per method; 60 rows (12 stress rows, 48 stochastic rows); metrics: attribution HHI, gross attribution, limited-mapping loss ratio, portfolio loss ratio, stress exposed weight share, stress NAV impact, technology-common loss ratio, top-family share.

Environment for both: Linux x86_64, CPU only; Python 3.12.3, NumPy 1.26.4, pandas 2.3.3, SciPy; RNG seed 20260714. Wall times: 3,706.476 s (Experiment I), 167.987 s (Experiment II).

B Artifact digests

SHA-256 manifest of the Experiment I pack:

366a483894354995e206bdecc1533476a99b47aad09906c16295868ff45a8218	README.md
8b5bcf99d39591c7ecb09ab02edceb5350cc3d83482333a22b799b6691d0d41a	SOURCE_ATTACHMENT_AUDIT.md
6e214e89ab769960e918b1e517ad58655a79913945e211b7d027757d64869c59	...stream_summary.json
2a860ff6be2363b718c93458569a1d4d5469bf0da36e847f5bd07c8adfa00902	...raw_metrics.json
bdca0e4e5fb6e08b4654f17ffc263dde2fa30e41fe774d34dbd2b2cd63cbc9d8	...stream_results.csv
2e6ea050f8af3b61e755e17d9e7ca4fda467235c583e5f5d28616a63af37aab6	run_progress.log
b09381e35b18a1fba718ab01ce90a6c38e1b14133f66f2fdde7e094cfd083eba	artifact_manifest.json

Runner scripts and input series (Experiment I):

63565f73f23a6ec0b73d11f1badf5478ca29e84b991e04923bb136fd4c2514f4	build script
ae9e3c75173088796dab4b0b56853ad0da7e1f977b486601a08a58cf8f396ef6	check script
a0a1acbc958f3a42feb9bc4269088ab8d7e4219922b5ced48abc7fd1107f765a	portfolio_return_series.csv

SHA-256 manifest of the Experiment II pack:

afc1fc34e38383f3fb38382abffeb0f8824120c7283bc1c39a7ddaa91f7e1071	README.md
d3fa5b45d11e3c44c00a8f2321ee529b8a6cc88397878ae814ff22db9ee55d2e	SOURCE_ATTACHMENT_AUDIT.md
8d556dc39132b2416646953e061421a0b68017040e4d3b81a3ff4b991eed6240	...raw_metrics.json
5779ee95d5aa66f7dfbb8942e9f906c9020155c4edd868a0521f9ee60b50d869	...stream_results.csv
038d45f98039f9cdf9a88231d5fed08f0c71f61a33d9a5ff5ee5df8a35db215f	...stream_summary.json
5eef9405744c44832bb5381aeb040c49013e70c110da20f3b1d5b8b0f412f5e5	run_progress.log

Runner scripts (Experiment II):

4ec8faa1738e2eddb1149a598067c29dd9a1500738dce1d440a30b170248814	build script
d4996244a4b0c7f52406e9b591ece2246badfaa342bda10bca75fcf7004b3c5d	check script